```
import tensorflow as tfprint(tf.__version__)
```

OUTPUT:
2.9.2

```
scalar = tf.constant(7)
print(scalar)
print(scalar.ndim)
```

OUTPUT:
tf.Tensor(7, shape=(), dtype=int32)
0

```
vector = tf.constant([10,10])
print(vector)
print(vector.ndim)
```

OUTPUT:
tf.Tensor([10 10], shape=(2,), dtype=int32)
1

```
matrix = tf.constant([[10,11],[12,13]])
print(matrix)
print(matrix.ndim)
```

OUTPUT:
tf.Tensor([[10 11][12 13]], shape=(2, 2), dtype=int32)
2

```
basic_tensor = tf.constant([[10,11],[12,13]])
print(basic_tensor)
```

OUTPUT:
tf.Tensor([[10 11][12 13]], shape=(2, 2), dtype=int32)

```
print(basic_tensor + 10)
print(basic_tensor - 10)
print(basic_tensor * 10)
print(basic_tensor / 10)
```

OUTPUT:
tf.Tensor([[20 21][22 23]], shape=(2, 2), dtype=int32)
tf.Tensor([[0 1][2 3]], shape=(2, 2), dtype=int32)
tf.Tensor([[100 110][120 130]], shape=(2, 2), dtype=int32)
tf.Tensor([[1. 1.1][1.2 1.3]], shape=(2, 2), dtype=float64)

```
tensor_011 = tf.constant([[2,2],[4,4]])
tensor_012 = tf.constant([[2,3],[4,5]])
```

```
print(tf.matmul(tensor_011,tensor_012))
```

OUTPUT:

```
tf.Tensor([[12 16][24 32]], shape=(2, 2), dtype=int32)
```

```
tensor_013 = tf.constant([[1,2,3],[4,5,6],[7,8,9]],dtype='float32')
print(tf.reduce_min(tensor_013))
print(tf.reduce_max(tensor_013))
print(tf.reduce_sum(tensor_013))
```

OUTPUT:

```
tf.Tensor(1.0, shape=(), dtype=float32)
tf.Tensor(9.0, shape=(), dtype=float32)
tf.Tensor(45.0, shape=(), dtype=float32)
```

```
print(tf.sqrt(tensor_013))
print(tf.square(tensor_013))
print(tf.math.log(tensor_013))
```

OUTPUT:

```
tf.Tensor([[1. 1.4142135 1.7320508][2. 2.236068 2.4494898][2.6457512 2.828427 3. ]],
shape=(3, 3), dtype=float32)
tf.Tensor([[ 1. 4. 9.][16. 25. 36.][49. 64. 81.]], shape=(3, 3), dtype=float32)
tf.Tensor([[0. 0.6931472 1.0986123][1.3862944 1.609438 1.7917595][1.9459102 2.0794415
2.1972246]], shape=(3, 3), dtype=float32)
```